2nd Town 22/11/24 M/S

USN | | | | | | | | | | |

[23TAIMLDE11]

THIRD SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

INTERNET OF THINGS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART – A			**5x2=10**
1.	**a.**	What is IoT? List out its Characteristics.	L1	CO1	**02**
	b.	What are the Fundamental components of IOT?	L1	CO2	**02**
	c.	Define Data Analytics.	L1	CO3	**02**
	d.	Write few Physical devices associated with IOT.	L1	CO4	**02**
	e.	Define Raspberry Pi.	L1	CO5	**02**
		PART - B			**6x5=30**
2.	**a.**	Write a note on Genesis of IoT.	L1	CO1	**05**
		OR			
	b.	Explain in detail about Convergence of IT and IoT.	L1	CO1	**05**
3.	**a.**	Write a short note on IoT Communication Models.	L2	CO2	**05**
		OR			
	b.	Write a short note on IoT Communication APIs.	L1	CO2	**05**
4.	**a.**	Explain How gateways give path to IoT Environment.	L1	CO3	**05**
		OR			
	b.	Discuss about device management and its advancement.	L1	CO3	**05**
5.	**a.**	How to configure a Raspberry Pi. Explain?	L1	CO4	**05**
		OR			
	b.	Write a brief note on Arduino UNO.	L2	CO4	**05**
6.	**a.**	Write in detail about Data representation and visualization.	L1	CO5	**05**
		OR			
	b.	Explain in detail about SSH and Remote access.	L1	CO5	**05**
7.	**a.**	Write a note on the drivers behind Network Architectures.	L1	CO1	**05**
		OR			
	b.	Explain in detail about implementing Industrial Automation by using IoT.	L1	CO5	**05**
		PART - C			**6x10=60**
8.	**a.**	Illustrate IoT Functional Stack with neat diagram and its Functions.	L1	CO1	**10**
		OR			
	b.	Explain in detail about IoT Data Management and Computation Stack.	L1	CO1	**10**

9.	**a.**	Write the main design principles and needed capabilities for IoT Architecture.	L1	CO2	**10**
		OR			
	b.	Outline a simplified IoT architecture. What are the key components and their roles?	L1	CO2	**10**
10.	**a.**	Explain briefly about the sensors and its Functionalities.	L2	CO3	**10**
		OR			
	b.	What do you mean by data management? How it Impacts IoT Architecture.	L1	CO3	**10**
11.	**a.**	Discuss the fundamentals of Arduino programming. What are the essential concepts and structures?	L1	CO4	**10**
		OR			
	b.	Discuss how to configure Raspberry Pi for IoT applications. What are the operating systems can be used?	L1	CO4	**10**
12.	**a.**	Explain in brief about the applications of IOT.	L2	CO5	**10**
		OR			
	b.	Write a case study for Commercial building automation.	L1	CO5	**10**
13.	**a.**	Write a short note on the Following. **i.** Physical Design of IoT **ii.** Logical Design of IoT	L1	CO2	**10**
		OR			
	b.	Write a short note on the Following. **i.** Wide area networking (WAN) **ii.** Local area networking (LAN)	L1	CO3	**10**